MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 09** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If PA and PB are tangents to a circle with centre O from point P such that $\angle AOB = 100°$, then $\angle APB$ is equal to: [1]

(A) $50°$ (B) $70°$ (C) $80°$ (D) $100°$

Q2. The incircle of $\triangle ABC$ touches sides BC, CA, AB at D, E, F. If $AB = 10$ cm, $BC = 8$ cm, $CA = 12$ cm, then length AF is: [1]

(A) 5 cm (B) 7 cm (C) 8 cm (D) 9 cm

Q3. A chord AB of a circle of radius 10 cm subtends a right angle at the centre O. Tangents at A and B intersect at P. The length of PA is: [1]

(A) 5 cm (B) 10 cm (C) $10\sqrt{2}$ cm (D) $5\sqrt{2}$ cm

Q4. Two concentric circles are of radii 15 cm and 9 cm. The length of the chord of the larger circle touching the smaller circle is: [1]

(A) 12 cm (B) 18 cm (C) 24 cm (D) 20 cm

Q5. How many tangents can be drawn to a circle from a point lying strictly inside the circle? [1]

(A) 0 (B) 1 (C) 2 (D) Infinitely many

Q6. **Assertion (A):** Tangents drawn at the end points of any diameter of a circle are parallel.
Reason (R): Radii to the points of contact are perpendicular to the respective tangents, and consecutive interior angles sum to 180°. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. If AB, AC, PQ are tangents to a circle touching at B, C, R respectively and $AB = 11$ cm, find the perimeter of $\triangle APQ$. [2]

Q8. In two concentric circles of radii 17 cm and 15 cm, find the length of the chord of the larger circle which is tangent to the smaller circle. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. In the given figure, OP is equal to the diameter of the circle with centre O. Tangents PA and PB are drawn. Prove that $\triangle PAB$ is an equilateral triangle. [3]

Q10. A quadrilateral $ABCD$ circumscribes a circle. Prove that $AB+CD = BC+DA$. If $AB = 6$ cm, $BC = 9$ cm, $CD = 8$ cm, find the perimeter of the quadrilateral. [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

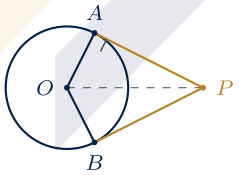
Q11. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that: (i) $\angle PTQ = 2\angle OPQ$, (ii) OT is the perpendicular bisector of chord PQ, (iii) If $\angle PTQ = 60^\circ$ and radius $OP = 6$ cm, find the length of OT. [5]

Q12. A right triangle ABC with sides $AB = 6$ cm, $BC = 8$ cm, and $\angle B = 90^\circ$ circumscribes a circle with centre O and radius r. (i) Find the inradius r. (ii) Find the lengths of the tangent segments AF, BD, CE. (iii) Prove that Area$(\triangle ABC) = r \cdot s$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Ferris Wheel Observation Platform and Tangent Safety Barriers

A giant observation wheel of radius $r = 12$ m is centred at O. From an elevated viewing platform P located 20 m from O ($OP = 20$ m), safety cable barriers PA and PB are strung tangentially to the wheel rim at points A and B.



(i) Calculate the length of the safety cable barrier PA. [1]

(ii) Find the measure of central angle $\angle AOB$ in terms of $\angle APB$. [1]

(iii) Calculate the area of quadrilateral $PAOB$, OR find the distance from O to the midpoint of chord AB. [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★